IT350 Data Analytics

Assignment 3

Team Members:

Rahul Saravanan (231IT054)
Aditi Pandey (231IT003)
Tejas Bajaj (231IT080)
Shishir Hegde (231IT067)

Individual Contributions :

Team 1 : Rahul and Aditi : Built and compared the performance of two primary neural network models: a Multi-Layer Perceptron (MLP) and a Long Short-Term Memory (LSTM) Network. They designed the MLP with a three-layer feed-forward architecture (128, 64, and 32 neurons) using ReLU activation and Adam optimization to map complex non-linear feature interactions. Additionally, they developed a sequential LSTM model with a 12-timestep lookback window, featuring dual layers of 64 and 32 units interleaved with Dropout to capture temporal dependencies.
Team 2 : Tejas and Shishir : Completed the assignment notebook, performing comprehensive data cleaning and Principal Component Analysis (PCA). They implemented and evaluated several similarity-based and predictive algorithms, including a K-Nearest Neighbors (KNN) Regressor and a Random Forest Regressor. Additionally, they conducted an in-depth analysis of distance metrics, specifically implementing Euclidean, Manhattan, and Cosine distances to understand data relationships and clustering behavior.

1. Introduction and Data Preprocessing

The objective of this analysis is to apply similarity-based algorithms and neural network models to understand patterns in energy consumption data. The dataset comprises temporal energy readings (e.g., active power, voltage, current) across multiple meters (labeled A through F) .

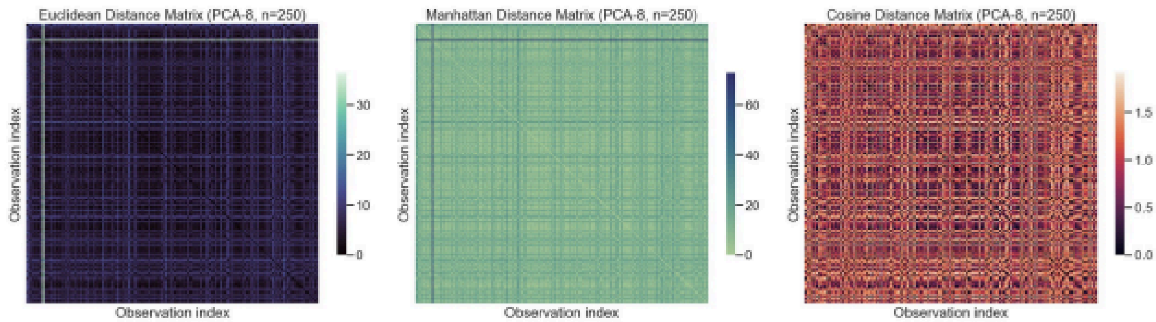
To prepare the data, we engineered features such as energy_intensity, apower_roll3, and temporal markers (is_peak, is_weekend) . After robust, standard, and min-max scaling , we applied Principal Component Analysis (PCA). Retaining 8 principal components captured 95.43% of the dataset's variance, significantly reducing dimensionality while preserving core patterns

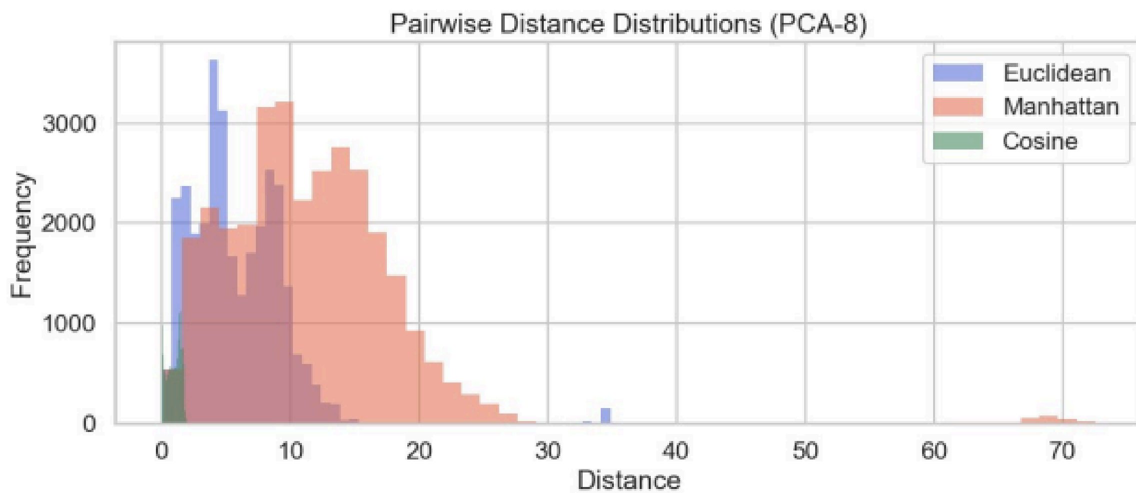
2. Similarity-Based Algorithms & Distance Measures

To analyze relationships between data points, we implemented three distinct distance metrics on a sample of the PCA-reduced space :

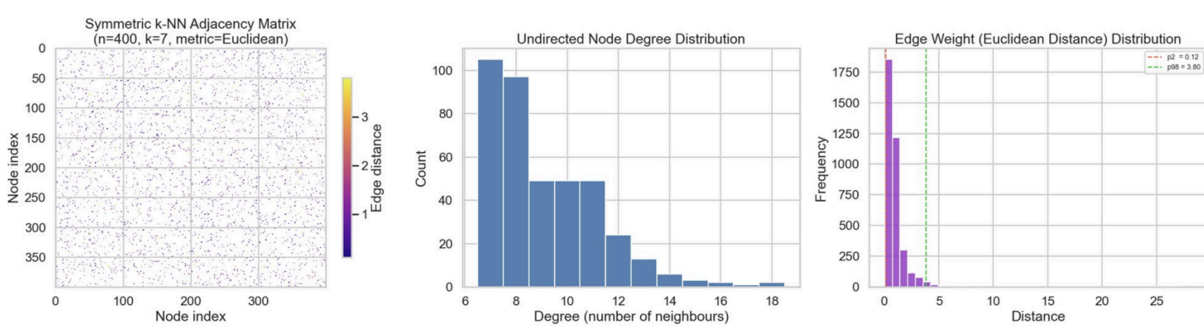
- **Euclidean Distance:** Captures the straight-line distance between energy readings.
- **Manhattan Distance:** Evaluates distance along axes, which is robust to outliers in power spikes.
- **Cosine Distance:** Measures the angle between feature vectors, effectively comparing the "shape" or profile of consumption regardless of total magnitude.

Observations: The pairwise distance distributions revealed that Cosine distances were heavily skewed toward 0, indicating strong directional alignment among many readings, whereas Euclidean and Manhattan distances showed a broader spread with distinct clusters representing varying power states.



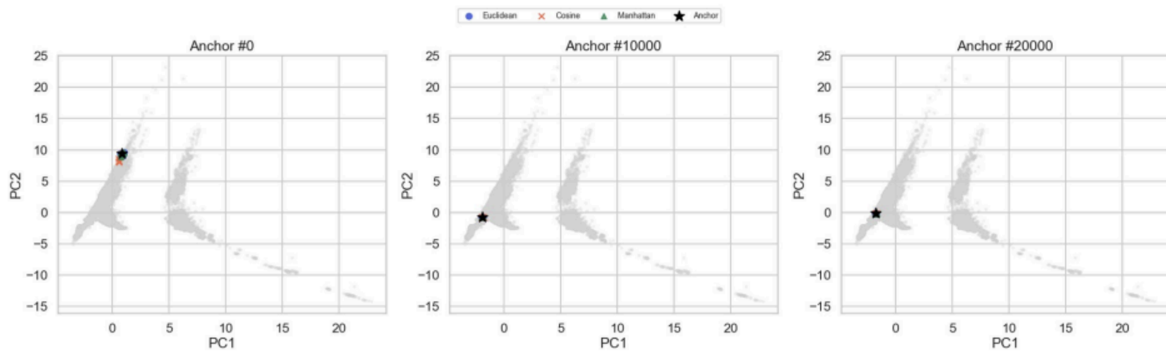


3. K-Nearest Neighbors (KNN) Search and Graph Representation



Unsupervised Anchor Search

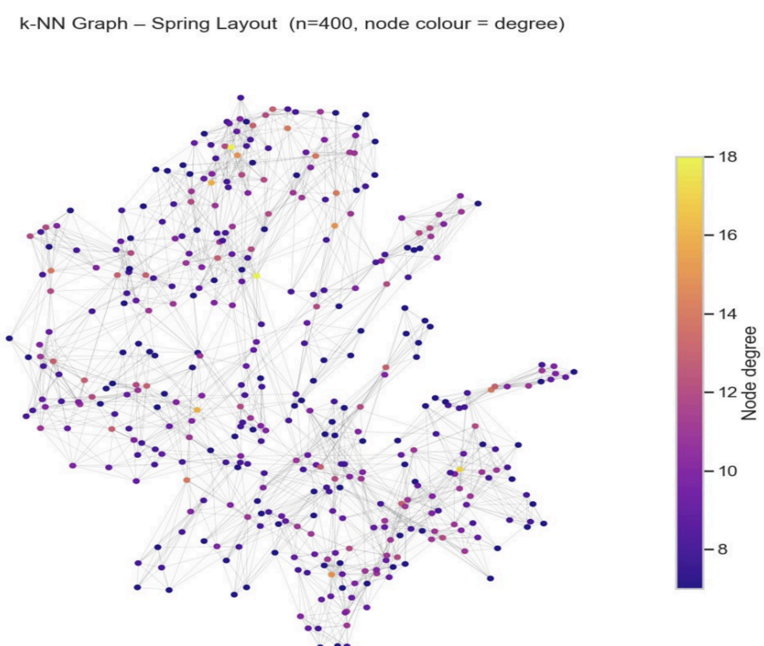
We utilized the NearestNeighbors algorithm (k=8) to identify the closest observations to specific "anchor" data points under different metrics . This allowed us to observe how different distance measures cluster the data physically (Euclidean) versus behaviorally (Cosine).



Graph Construction

To visualize the interconnectedness of the data, we constructed a symmetric k-NN adjacency matrix using k=7 and the Euclidean metric.

Visualizations & Insights: The NetworkX spring-layout visualization of the graph clustered nodes based on their Euclidean similarity . Nodes with higher degrees (colored differently) acted as hubs, representing common operational states of the meters .



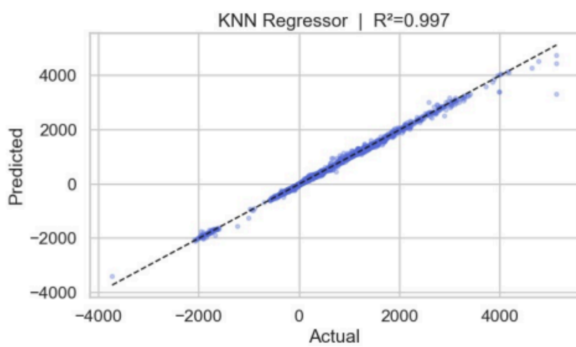
4. Predictive Modeling: KNN and Neural Networks

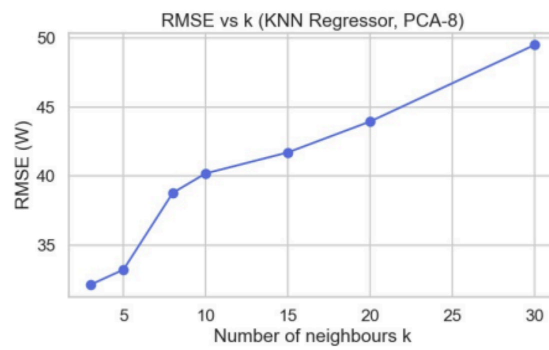
To predict active power (apower_ph1), we tested similarity-based regression and two deep learning architectures.

A. K-Nearest Neighbors Regressor

We implemented a KNeighborsRegressor using distance weighting and Minkowski geometry. A sweep of k values demonstrated that prediction error (RMSE) increased as k expanded from 3 to 30 .

- **Performance:** MAE: 12.54 W | RMSE: 40.18 W | R^2: 0.997.

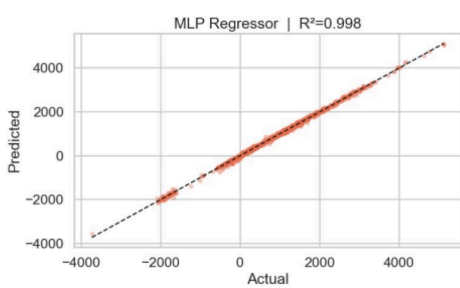


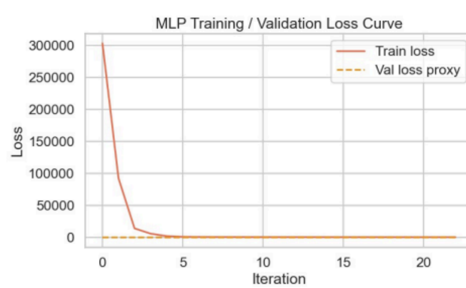


B. Multi-Layer Perceptron (MLP)

We designed a feed-forward MLP with three hidden layers (128, 64, 32 neurons) utilizing ReLU activation and Adam optimization . Early stopping was applied to prevent overfitting.

- **Performance:** MAE: 16.37 W | RMSE: 27.03 W | R^2: 0.998.
- **Insight:** The MLP outperformed the KNN regressor in minimizing large errors (lower RMSE), demonstrating its ability to map complex, non-linear feature interactions mapped by the PCA components.

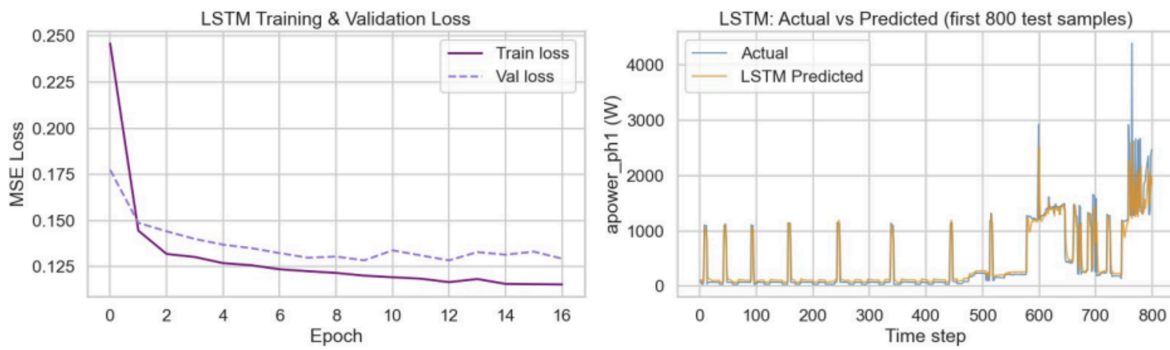




C. Long Short-Term Memory (LSTM) Network

Given the temporal nature of the data, we restructured the dataset into sequences with a lookback window of 12 timesteps. We implemented an RNN architecture featuring two LSTM layers (64 and 32 units) interleaved with Dropout to handle sequential dependencies .

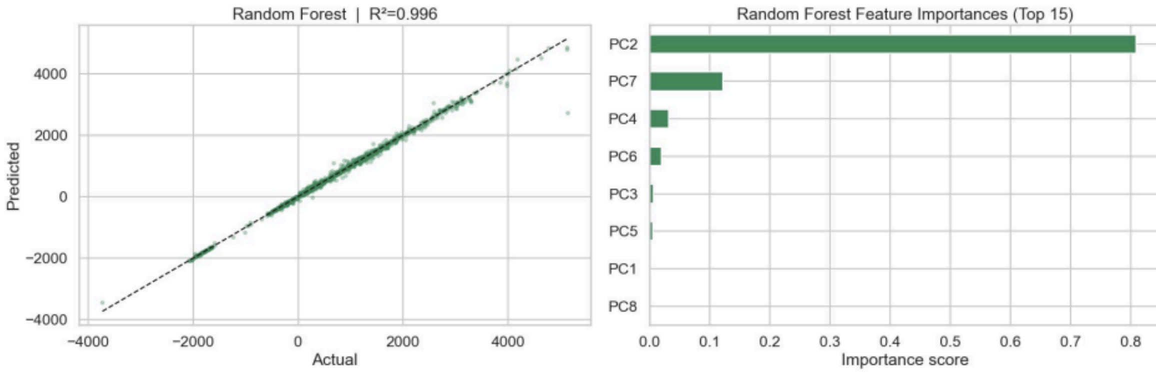
- **Performance:** MAE: 112.97 W | RMSE: 252.81 W | R^2: 0.805.
- **Insight:** The LSTM struggled compared to the MLP. This indicates that for this specific target, the engineered global features (like apower_roll3 and time-of-day indicators) passed through PCA provided a stronger predictive signal to the MLP than the raw temporal sequences provided to the LSTM.

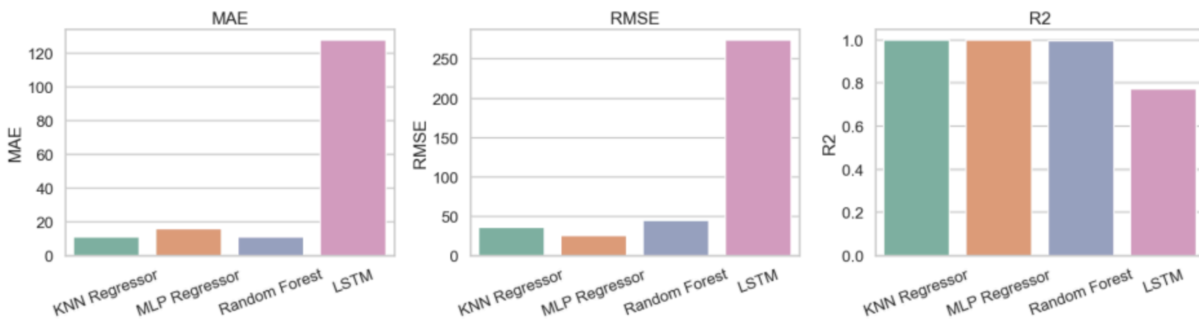


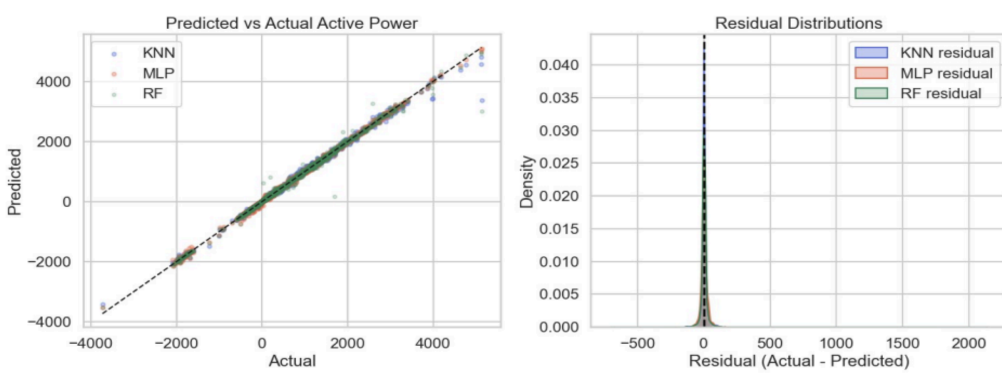
D. Random Forest Regressor

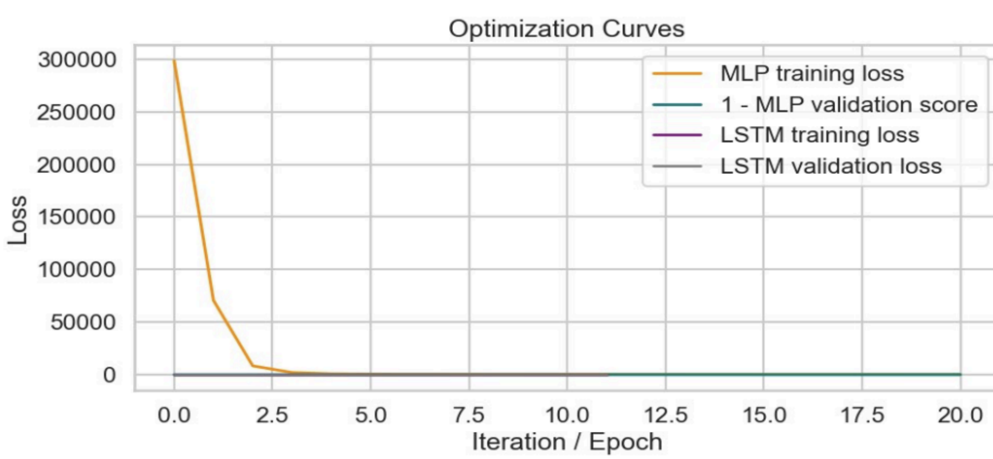
To establish a robust, tree-based ensemble baseline, we implemented a RandomForestRegressor configured with 300 estimators, a maximum depth of 16, and a minimum of 2 samples per leaf .

- **Performance:** MAE: 11.82 W | RMSE: 42.37 W | R^2: 0.996.
- **Insight:** The Random Forest model performed exceptionally well, achieving an R^2 score highly comparable to the KNN regressor and slightly trailing the MLP. More importantly, the model inherently allowed us to calculate feature importances. The analysis revealed that Principal Component 2 (PC2) overwhelmingly drove the model's predictions, followed distantly by PC7 and PC4 . This demonstrates that the specific variance captured by PC2 during our dimensionality reduction contains the most critical mathematical signature for determining a meter's active power.


Random Forest | R²=0.996
Predicted
Actual
Random Forest Feature Importances (Top 15)
PC2
PC7
PC4
PC6
PC3
PC5
PC1
PC8
Importance score


MAE
RMSE
R2
KNN Regressor
MLP Regressor
Random Forest
LSTM

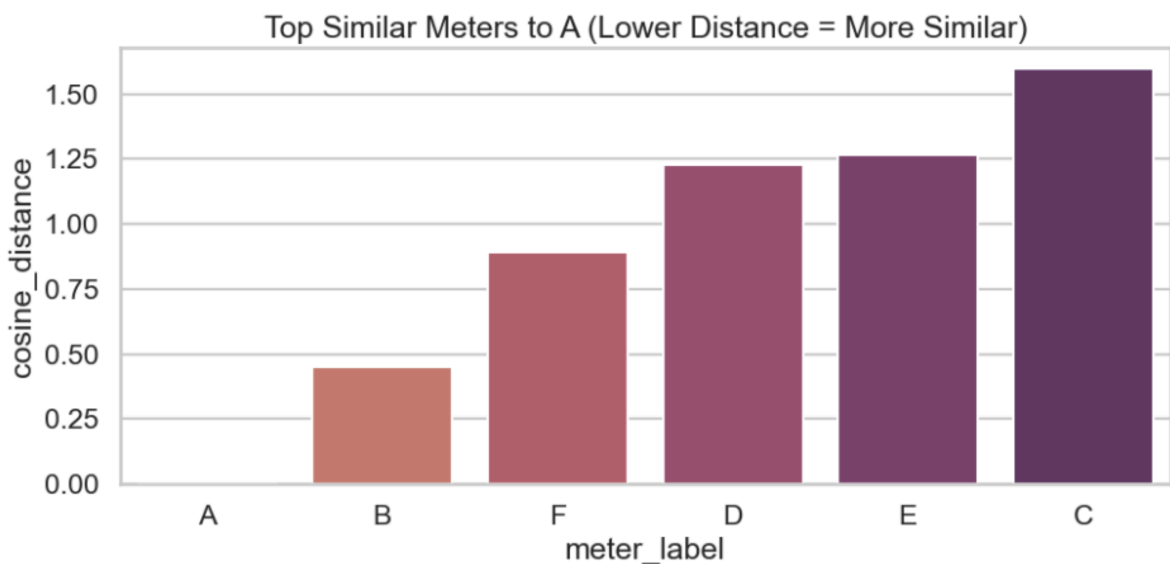

Predicted vs Actual Active Power
KNN
MLP
RF
Predicted
Actual
Residual Distributions
KNN residual
MLP residual
RF residual
Density
Residual (Actual - Predicted)


Optimization Curves
MLP training loss
1 - MLP validation score
LSTM training loss
LSTM validation loss
Loss
Iteration / Epoch

5. System-Wide Profiling: Meter Similarity

Beyond predicting power, we utilized similarity metrics to profile the physical meters. By aggregating median features for each meter and computing pairwise Cosine distances, we identified operational similarities.

- **Results:** Meter B is the most functionally similar to Meter A (Cosine Distance ≈ 0.45), while Meter C exhibits an entirely different consumption profile (Distance ≈ 1.59). This proves similarity metrics can successfully cluster hardware based on usage signatures.



6. Applications in Large-Scale Machine Learning and Research Trends

The techniques implemented in this assignment mirror current research trends in large-scale energy analytics:

1. **Smart Grid Management:** Graph representations (like our k-NN graph) are increasingly used in Graph Neural Networks (GNNs) to model the topology of power grids, enabling real-time load balancing and anomaly detection.
2. **Predictive Maintenance:** Similarity search (KNN) combined with Cosine distance is used at scale to match live sensor readings against historical failure signatures. When a meter's profile suddenly shifts its "nearest neighbors" away from healthy clusters, it triggers maintenance alerts.
3. **Time-Series Forecasting at Scale:** While our MLP outperformed the LSTM here, modern large-scale forecasting relies heavily on transformer-based architectures and

advanced RNNs to predict peak demand across millions of nodes simultaneously, optimizing energy distribution.

7. Conclusion

This project successfully demonstrated the utility of similarity-based algorithms and neural networks in data analytics. Distance measures (Euclidean, Cosine) proved highly effective for both localized prediction (KNN Regressor) and global profiling (Meter-to-Meter similarity). Furthermore, comparing neural network architectures highlighted that while LSTMs are theoretically suited for sequential data, a well-engineered feature set paired with a robust MLP can yield superior predictive accuracy ($R^2 = 0.998$) with significantly lower computational overhead.